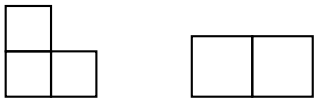
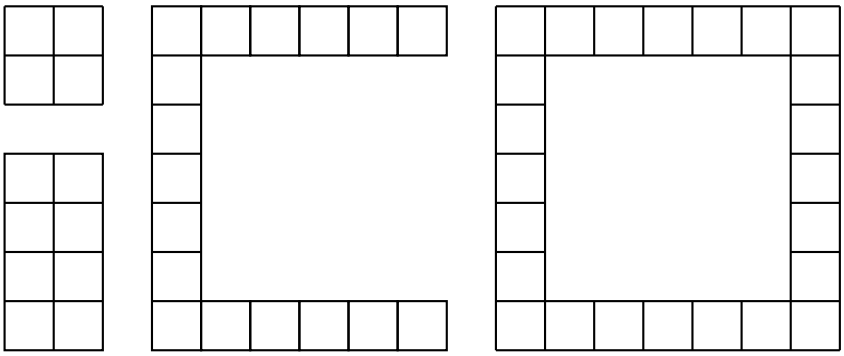
Combinatorics Olympiad
Junior Level
October 2025

Problems

1. The domino and L-tromino pieces are as shown below.

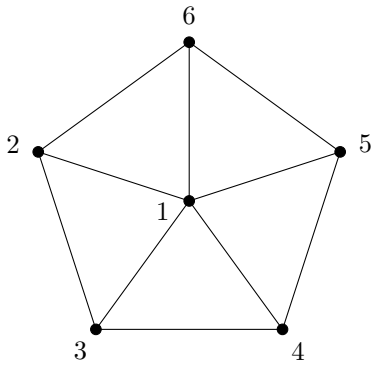


In how many ways can we tile the given shape shown in the figure below using some number of dominoes and L-trominoes, with no overlaps?
Note: rotations of pieces are allowed.

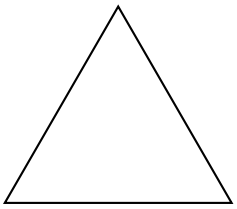


2. 16 people are sitting around a circle. Two people are called friends if at least 4 people are sitting between them in each of the two arcs connecting them around the circle. In how many ways can we choose three people such that any two of these three people are friends?
Note: The order of selection does not matter.

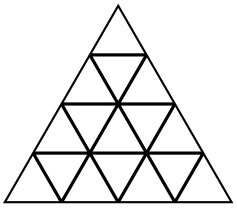
3. In how many ways can we color some of the segments in the following figure red so that each of the six points is connected to an odd number of red segments?



4. We have a "triangle-drawing machine". Using this machine at each step we can draw one triangle on the plane. Initially, one equilateral triangle is drawn on the plane.



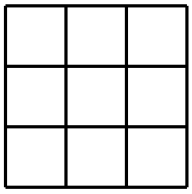
We want to transform it into the following shape:



What is the minimum number of steps we need to obtain this shape?

5. We have a "rectangle-drawing machine". Using this machine at each step we can draw one rectangle on the plane. In how many ways can we draw a 3×3 table in exactly 3 steps using the rectangle-drawing machine?

Note: Two methods that draw the same set of rectangles in a different order are considered the same.

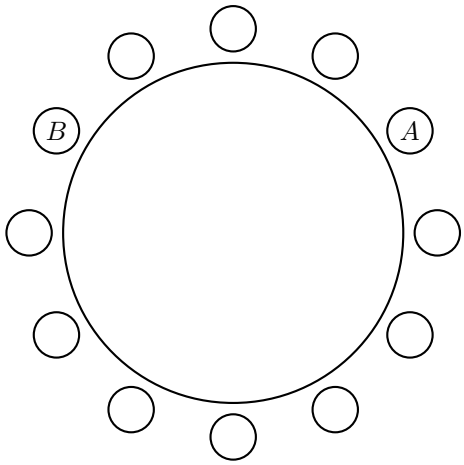


6. Find the number of positive integers with at most 11 digits that are not divisible by 10 but are divisible by their rightmost digit.

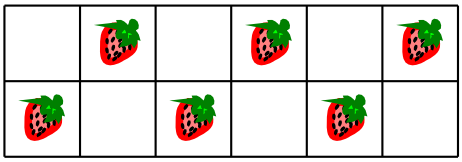
7. 117 gangsters are standing in a line. Simultaneously, each gangster shoots the nearest person to himself. (If two people are equally close — one on each side — the gangster may choose either of them.) What is the minimum possible number of gangsters who get shot?

8. 12 classmates stand around a circle in a fixed order. A natural number d with $d < 12$ is called *interesting* if it is possible to partition the 12 classmates into 6 two-person teams so that, in each team, the two teammates have $d - 1$ people between them along the circle (for example, in the figure below, the number of people between A and B can be considered as 3 or 7).

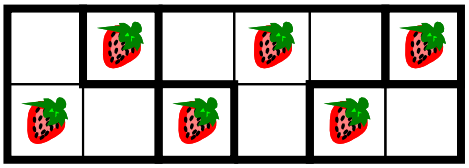
   How many interesting numbers d are there?



9. A 2×6 table is given, and in some of its cells, strawberries are placed as shown below. In how many ways can we divide this table along the table lines into 6 parts such that each part contains exactly one strawberry?



   An example of a valid partition is shown in the figure below:



10. 10 uncolored balls with numbers 1 to 10 are arranged around a circle. In how many ways can we color each ball by red or blue so that (1) there are exactly 5 blue and 5 red balls, and (2) for each red ball, among the two balls immediately to its left and the two balls immediately to its right, there are at least two red balls in total?

11. Dara and Sara play the following game:

    There are 8 points labeled 1 through 8 on the number line. Initially, all points are uncolored. In each turn, Dara chooses one of the uncolored points, and Sara colors it either black or white.

    After each turn, if it is possible to draw a straight line such that all white points lie on one side of the line and all black points lie on the other side of the line, Dara earns one coin. (Indeed, if all the colored points so far are of the same color, Dara earns one coin.)

    Dara wants to maximize the number of coins that he earns, and Sara wants to minimize that. Assuming both play optimally, find the number of coins that Dara has at the end of the game.

12. Eight teams, numbered 1 to 8, participate in a single-elimination soccer tournament.

    In a match between host a and guest b:

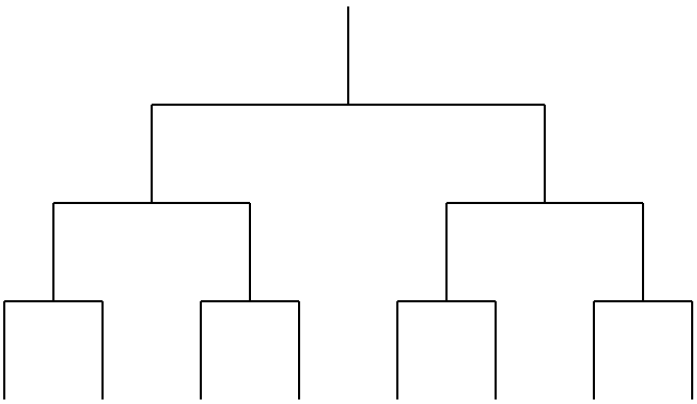
    - If $a < b$, the host wins.
    - If $a > 2b$, the guest wins.
    - In all other cases, either team may win.

    The tournament is held over three rounds to determine the champion. In each round, the teams are paired into matches, with one team serving as the host.

    The hosting rule is as follows: If one of the two teams has hosted fewer matches so far, that team becomes the host. If both have hosted the same number of matches, the host is chosen randomly.

    In how many distinct possible tournament scenarios can team number 8 become the champion?

    Note: Two tournament scenarios are considered different if there exists a round in which a certain match is played in one but not in the other, or if the same match occurs in both but the host team differs in the two cases.



13. The education system of a country consists of exactly five subjects. Each school must schedule these five subjects during the first five days of the week (Monday to Friday), exactly one subject per day.

    The school employs five teachers (one per subject). Each teacher proposes two possible days of the week on which they can teach their subject as their availabilities.

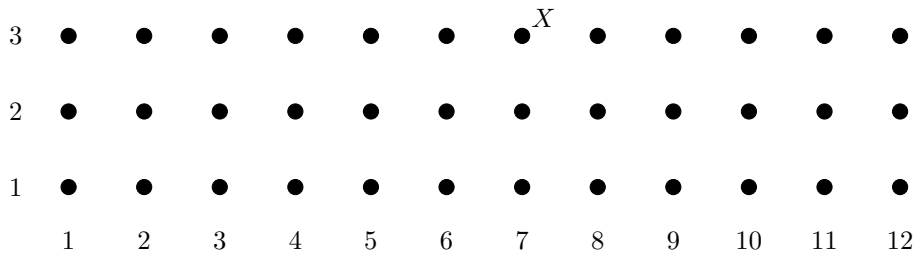
    Suppose that based on the proposed days by the teachers, the school can schedule the weekly program in x ways. What is the maximum possible value of x?

14. The houses of a village are arranged on the points of a 3×12 rectangular grid, as shown in the figure below. That is, the village has 36 houses, one at each grid point. A house located in row i and column j of the grid is labeled (i, j). For instance, the house marked by X is labeled $(3, 7)$.

    Traveling from house (i, j) to house (i', j') using jet takes a number of minutes equal to the difference in their row numbers plus the difference in their column numbers. For example, traveling from house $(2, 2)$ to $(1, 4)$ takes 3 minutes.

    A family living in $(1, 1)$ wants to visit all the other houses (each house exactly once), and then return to their own house. They use jet for traveling between houses.

    Find the minimum possible total time required for this process.



15. Hamid's Grandpa has passed away, and his old suitcase has been inherited by Hamid. The lock on this suitcase is a two-digit display, initially showing 00. For this lock, there is a unique two-digit pass-code. As soon as the correct pass-code appears on the display, the lock opens.

    There are four buttons:

    - Button 1 increases the right digit by 1.
    - Button 2 decreases the right digit by 1.
    - Button 3 increases the left digit by 1.
    - Button 4 decreases the left digit by 1.

    A digit cannot be increased if it is already 9, and cannot be decreased if it is already 0.

    Hamid does **not** know the correct pass-code, but he remembers his grandfather saying that (1) neither 09 nor 90 is the lock pass-code, and (2) if the display ever shows 09 or 90, the suitcase will be locked forever.

    What is the minimum number of times that Hamid must press the buttons to ensure that he will definitely open the lock, regardless of what the correct pass-code is?